

EMRE BATU ATAKAN

emrebatuatakan.github.io | e.batuatakan@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Waterloo

Waterloo, Ontario, Canada | 2025 - 2030

Bachelor of Mathematics in Computer Science (Honours) with Co-op, GPA: 3.6/4.0

- Courses: Designing Functional Programs (Advanced Level), Calculus 2, Linear Algebra 1

Awards

- **ChessHacks Waterloo 2025:** Pufferfish Chess Engine, **3rd Place**
- **American Mathematics Olympiad 2024-2025:** Silver Medal (National 7th / Global 269th) 2024, Bronze Medal 2025
- **International STEM Olympiad:** Mathematics Gold Medal, Global 12th (2024), Global 14th (2023)
- **UWaterloo Math Contests:** Fermat 2024 Honor Roll, Distinction; Cayley 2023 Distinction

EXPERIENCE

Shopify Inc. — Software Engineering Intern

Toronto, Winter 2026

Automated Shop Analysis Pipeline - A distributed pipeline that analyzes and scores e-commerce shops for fraud, deception, and business legitimacy to make the platform more secure for users.

- Built the full pipeline prototype from scratch: reads shops from BigQuery, uses dbt to identify new and stale shops needing analysis, distributes processing through Apache Flink with a custom Java UDF calling a FastAPI ML service, and writes results back to BigQuery, orchestrated by an Airflow DAG for daily automated runs across **800K+ shops**, with **100K+ shops** processed on daily incremental runs.
- Ported the prototype to the Shopify monorepo, implementing three AI subagents (fraud detection, business profiling, merchant classification), an orchestrator with parallel execution, and **129** unit tests.
- Developed an async HTTP pre-filter using aiohttp with **500 concurrent requests** to skip inaccessible URLs before model processing, reducing wasted GPU compute on blocked or unreachable sites.
- Built benchmarking infrastructure tracking throughput, success rates, and P50/P95/P99 latency metrics. Used to optimize concurrency settings across pipeline stages.

AI Slackbot Evaluation Framework & Analytics Dashboard

- Designed and built an A/B testing framework for comparing AI Slackbot skill configurations side-by-side. The framework runs the same prompts against different configurations, scores them with LLM judges and deterministic assertions, and publishes structured results to a BigQuery schema for full reproducibility.
- Built an analytics dashboard visualizing Slackbot usage data from BigQuery, including deep link filtering, keyboard navigation, insights with leaderboards, GitHub-style activity heatmaps, and pattern analytics (time-of-day activity, tool error clustering, channel leaderboards).
- Fixed critical auth and publishing issues blocking evals from running inside production containers, unblocking the team to run evals directly in production-like environments.

University of Waterloo — Robotics Research

June 2026 - Present

- Leading 8-person research team developing end-to-end visuomotor policies. Mentored by [Generalist](#)'s Research Scientist.
- Designed and executed controlled experiments comparing DINOv3-Transformer, CNN/MLP, and pretrained ConvNeXt behavior-cloning policies, analyzing the gap between offline training loss and closed-loop task performance.
- Ran hyperparameter sweeps over observation windows, action-history lengths, action-chunk horizons, auxiliary localization losses, dataset variants, and random seeds; tracked experiments in W&B and evaluated checkpoints on identical seeded rollouts.
- Identified that the larger Transformer policy was memorizing demonstrations and generalizing poorly, motivating a smaller CNN/MLP policy that improved randomized pick-and-place success from approximately **30% to 99.8%**.
- Validated the selected policy over **2,000+** randomized closed-loop rollouts during a multi-hour live demonstration, observing only **four** failures and confirming readiness for hardware deployment.
- Integrated trained checkpoints into the asynchronous inference loop and tested interactions between model latency, observation sampling, replanning, and 100 Hz command execution on commodity hardware.
- Operated parallel data-collection jobs and curated 13 hours of synchronized pick-and-place demonstrations; launched and monitored SLURM training and evaluation workloads on Digital Research Alliance of Canada A100/H100 systems.

Thresh Power — Software Development Intern

Summer 2024

Thresh Power is a San Francisco startup that streamlines M&A processes using machine learning.

- Worked on OpenAI API integration for document summarization and Q&A. Developed benchmarks to compare Large Language Model (LLM) performance. Built full-stack product features.
- Built and deployed a Python microservice on GCP. The service uses the OpenAI Assistant API to generate due diligence reports on user-provided files. Implemented separate staging and production environments for safer testing.
- Evaluated Ragas and various metrics as a means of understanding the performance impact of different prompting strategies. Documented findings and criticisms of different methodologies.
- Shipped a streaming file downloading feature as a React component with a corresponding Python API. Implemented 100% unit test coverage for the Python API. The feature is used by customers every day.

PROJECTS

Pufferfish

2025-2026

- Built Pufferfish, a chess engine/bot that combines a PyTorch-trained NNUE evaluation network with iterative-deepening alpha-beta search, quiescence search, transposition tables, move ordering heuristics, and time-managed decision logic. Developed the full pipeline from FEN feature encoding and large-scale training workflows to FastAPI-based inference integration for ChessHacks, with Modal-powered GPU training and a parallel C++ engine prototype for search and runtime optimization. Placed 3rd at ChessHacks Waterloo 2025.

Speech to Phonemes

Summer 2025

- Created a custom 120-hour dataset of high-quality sound-to-phoneme labels and integrated the new dataset with the dataloader's dataset mixing.
- Implemented an optimized multi-process pipeline in Python.
 - Batch GPU inference on Whisper to localize word timestamps; optimized GPU utilization by using a dataloader and pinned memory.
 - Used OpenAI API with multithreading to build phonemes for out-of-vocabulary words and expanded CMUDict by adding 4000 new word phonemes.
 - Implemented a dynamic programming algorithm to optimize for the best word-to-phoneme alignment globally.
- Ran A100 training jobs on the new dataset and configured hyperparameters accordingly.

LEADERSHIP

FRC Team Captain of RedFox 10247 (2025): Co-founded the team, led 26 members, raised **\$10,000+** in sponsorships, and trained 23 newcomers through custom robot-control simulations and hands-on workshops in Java, electronics, and mechanical design. Placed 11th at regionals and captained Alliance 7 as a rookie team.

FRC Team Captain of The Chaotics 8795 (2023-2024): Led 122 members across robotics and public relations, raised **\$10,000+**, and built out a CNC-equipped workshop from an unused school basement.

SKILLS

Languages: Python, Racket/Lisp, C, C++

Technical Skills: Machine Learning, Robotics, PyTorch, ROS, FastAPI, BigQuery, Apache Flink, dbt, Airflow, Docker, React, GCP, SolidWorks